

Модуль 4. Практика 3.

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1)

```
Switch#show vlan
Switch#show vlan
```

VLAN	Name	Status	Ports
1	default	active	Gi2/0
20	VLAN20	active	
100	VLAN100	active	
200	VLAN0200	active	
300	VLAN0300	active	
333	VLAN333	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

VLAN 20 и 333 существуют, но пусты.

Настройка интерфейсов на первом маршрутизаторе

```
Switch(config)#int range gi0/3,gi1/1,gi1/3
Switch(config-if-range)#switchport trunk enc dot1
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#switchport trunk native vlan 333
Switch(config-if-range)#switchport trunk allow vlan 333
```

```
Switch(config)#int range gi0/2,gi1/0,gi1/2
Switch(config-if-range)#switchport trunk enc dot1
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#switchport trunk allow vlan 20
```

show int trunk

Switch#show int trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	desirable	n-isl	trunking	1
Gi0/1	desirable	n-isl	trunking	1
Gi0/2	on	802.lq	trunking	1
Gi0/3	on	802.lq	trunking	333
Gi1/0	on	802.lq	trunking	1
Gi1/1	on	802.lq	trunking	333
Gi1/2	on	802.lq	trunking	1
Gi1/3	on	802.lq	trunking	333

Port Vlan allowed on trunk

Gi0/0	1-4094
Gi0/1	1-4094
Gi0/2	20
Gi0/3	333
Gi1/0	20
Gi1/1	333
Gi1/2	20
Gi1/3	333

Port Vlan allowed and active in management domain

Gi0/0	1,20,100,200,300,333
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Port Vlan allowed and active in management domain

Gi0/1	1,20,100,200,300,333
Gi0/2	20
Gi0/3	333
Gi1/0	20
Gi1/1	333
Gi1/2	20
Gi1/3	333

Port Vlan in spanning tree forwarding state and not pruned

Gi0/0	1,20,100,200,300,333
Gi0/1	20,100,200,300,333
Gi0/2	20
Gi0/3	333
Gi1/0	20
Gi1/1	333
Gi1/2	20
Gi1/3	333

Switch#

Настройка интерфейсов на коммутаторах 3-5

```
Switch(config)#int range g0/0-1
Switch(config-if-range)#switchport trunk enc dot1
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#exit
Switch(config)#int g0/0
Switch(config-if)#switchport trunk allow vlan 20
Switch(config-if)#exit
Switch(config)#int g0/1
Switch(config-if)#switchport trunk native vlan 333
Switch(config-if)#switchport trunk allow vlan 333
Switch(config-if)#exit
Switch(config)#int g1/0
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#int g1/1
Switch(config-if)#switchport access vlan 333
```

sh vlan br

sh int tr

* type "show ?" for a list of subcommands

Switch#show vlan br

VLAN	Name	Status	Ports
1	default	active	
20	VLAN20	active	Gi1/0
100	VLAN100	active	
200	VLAN0200	active	
300	VLAN0300	active	
333	VLAN333	active	Gi1/1
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

Switch#

Switch#show int tr

Switch#show int trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Gi0/1	on	802.1q	trunking	333
Gi0/2	desirable	n-isl	trunking	1
Gi0/3	desirable	n-isl	trunking	1

Port	Vlans allowed on trunk
Gi0/0	20
Gi0/1	333
Gi0/2	1-4094
Gi0/3	1-4094

Port	Vlans allowed and active in management domain
Gi0/0	20
Gi0/1	333
Gi0/2	1,20,100,200,300,333
Gi0/3	1,20,100,200,300,333

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	20
Gi0/1	333
Gi0/2	1,20,100,200,300,333
Gi0/3	20,100,200,300,333

Switch#

2)

```
PC1> ip 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0

PC1> ping 192.168.1.2

host (192.168.1.2) not reachable

PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=6.233 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=4.497 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=3.526 ms
^C
PC1> ping 192.168.1.4

host (192.168.1.4) not reachable

PC1> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=4.043 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=9.889 ms
^C
PC1> ping 192.168.1.6

host (192.168.1.6) not reachable

PC1> save
Saving startup configuration to startup.vpc
. done
```

```
PC2> ip 192.168.1.2
Checking for duplicate address...
PC2 : 192.168.1.2 255.255.255.0

PC2> ping 192.168.1.1

host (192.168.1.1) not reachable

PC2> ping 192.168.1.3

host (192.168.1.3) not reachable

PC2> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=2.214 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=2.129 ms
^C
PC2> ping 192.168.1.5

host (192.168.1.5) not reachable

PC2> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=5.474 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=5.711 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=4.991 ms
^C
PC2> save
Saving startup configuration to startup.vpc
. done

PC2> █
```

```
PC3> ip 192.168.1.3
Checking for duplicate address...
PC3 : 192.168.1.3 255.255.255.0

PC3> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=7.039 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=12.613 ms
^C
PC3> ping 192.168.1.2

host (192.168.1.2) not reachable

PC3> save
Saving startup configuration to startup.vpc
. done

PC3>
PC3> ping 192.168.1.4

host (192.168.1.4) not reachable

PC3> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=13.983 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=8.586 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=15.900 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=2.447 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=4.794 ms
^C
PC3> ping 192.168.1.6

host (192.168.1.6) not reachable
```

3)

Теги содержатся в пакетах передаваемых в VLAN 20

(отслеживается trunk линк, сам тег выделен)

No.	Time	Source	Destination	Protocol	Length	Info
41	48.023674	Private_66:68:02	Broadcast	ARP	68	Who has 192.168.1.1? Tell 192.168.1.3
42	48.027248	Private_66:68:00	Private_66:68:02	ARP	68	192.168.1.1 is at 00:50:79:66:68:00
→	43	48.030412	192.168.1.3	192.168.1.1	ICMP	102 Echo (ping) request id=0x714b, seq=1/256, ttl=64
←	44	48.035372	192.168.1.1	192.168.1.3	ICMP	102 Echo (ping) reply id=0x714b, seq=1/256, ttl=64
	46	49.038032	192.168.1.3	192.168.1.1	ICMP	102 Echo (ping) request id=0x724b, seq=2/512, ttl=64
	47	49.040978	192.168.1.1	192.168.1.3	ICMP	102 Echo (ping) reply id=0x724b, seq=2/512, ttl=64
	48	50.053883	192.168.1.3	192.168.1.1	ICMP	102 Echo (ping) request id=0x734b, seq=3/768, ttl=64
	49	50.059873	192.168.1.1	192.168.1.3	ICMP	102 Echo (ping) reply id=0x734b, seq=3/768, ttl=64
	51	51.062601	192.168.1.3	192.168.1.1	ICMP	102 Echo (ping) request id=0x744b, seq=4/1024, ttl=64
	52	51.070900	192.168.1.1	192.168.1.3	ICMP	102 Echo (ping) reply id=0x744b, seq=4/1024, ttl=64
	53	52.073609	192.168.1.3	192.168.1.1	ICMP	102 Echo (ping) request id=0x754b, seq=5/1280, ttl=64
	54	52.078577	192.168.1.1	192.168.1.3	ICMP	102 Echo (ping) reply id=0x754b, seq=5/1280, ttl=64

▶ Frame 43: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0

▶ Ethernet II, Src: Private_66:68:02 (00:50:79:66:68:02), Dst: Private_66:68:00 (00:50:79:66:68:00)

▼ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20

000. = Priority: Best Effort (default) (0)

...0 = DEI: Ineligible

.... 0000 0001 0100 = ID: 20

Type: IPv4 (0x0800)

▶ Internet Protocol Version 4, Src: 192.168.1.3, Dst: 192.168.1.1

▶ Internet Control Message Protocol

Тегов не будет в случае access портов и native trunk:

Захват с Standard input [Layer2Switch-1 Ethernet3 to Layer2Switch-3 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

arp || icmp

No.	Time	Source	Destination	Protocol	Length	Info
121	148.104650	Private_66:68:03	Broadcast	ARP	64	Who has 192.168.1.2? Tell 192.168.1.4
122	148.106640	Private_66:68:01	Private_66:68:03	ARP	64	192.168.1.2 is at 00:50:79:66:68:01
123	148.110192	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) request id=0xd94b, seq=1/256, ttl=
124	148.111992	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) reply id=0xd94b, seq=1/256, ttl=
126	149.116121	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) request id=0xda4b, seq=2/512, ttl=
127	149.118909	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) reply id=0xda4b, seq=2/512, ttl=
129	150.125442	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) request id=0xdb4b, seq=3/768, ttl=
130	150.134787	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) reply id=0xdb4b, seq=3/768, ttl=
134	151.144269	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) request id=0xdc4b, seq=4/1024, ttl=
135	151.150725	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) reply id=0xdc4b, seq=4/1024, ttl=
136	152.152984	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) request id=0xdd4b, seq=5/1280, ttl=
137	152.158468	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) reply id=0xdd4b, seq=5/1280, ttl=

Frame 123: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
Ethernet II, Src: Private_66:68:03 (00:50:79:66:68:03), Dst: Private_66:68:01 (00:50:79:66:68:01)
Internet Protocol Version 4, Src: 192.168.1.4, Dst: 192.168.1.2
Internet Control Message Protocol

Native trunk

Захват с Standard input [Layer2Switch-3 Ethernet4 to PC1 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

arp || icmp

No.	Time	Source	Destination	Protocol	Length	Info
27	38.016565	Private_66:68:02	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.3
28	38.016743	Private_66:68:00	Private_66:68:02	ARP	64	192.168.1.1 is at 00:50:79:66:68:00
29	38.023187	192.168.1.3	192.168.1.1	ICMP	98	Echo (ping) request id=0x714b, seq=1/256, ttl=6
30	38.023343	192.168.1.1	192.168.1.3	ICMP	98	Echo (ping) reply id=0x714b, seq=1/256, ttl=6
32	39.031380	192.168.1.3	192.168.1.1	ICMP	98	Echo (ping) request id=0x724b, seq=2/512, ttl=6
33	39.031568	192.168.1.1	192.168.1.3	ICMP	98	Echo (ping) reply id=0x724b, seq=2/512, ttl=6
36	40.046594	192.168.1.3	192.168.1.1	ICMP	98	Echo (ping) request id=0x734b, seq=3/768, ttl=6
37	40.046782	192.168.1.1	192.168.1.3	ICMP	98	Echo (ping) reply id=0x734b, seq=3/768, ttl=6
39	41.055217	192.168.1.3	192.168.1.1	ICMP	98	Echo (ping) request id=0x744b, seq=4/1024, ttl=
40	41.055440	192.168.1.1	192.168.1.3	ICMP	98	Echo (ping) reply id=0x744b, seq=4/1024, ttl=
41	42.066425	192.168.1.3	192.168.1.1	ICMP	98	Echo (ping) request id=0x754b, seq=5/1280, ttl=
42	42.066617	192.168.1.1	192.168.1.3	ICMP	98	Echo (ping) reply id=0x754b, seq=5/1280, ttl=

Frame 30: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:02 (00:50:79:66:68:02)
Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.3
Internet Control Message Protocol

access nopr

4)